

Thangatamil A

Software Development Engineer

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PROFILE

Software Development Engineer with 2.5 years of hands-on experience in orchestrating, designing and developing software solutions. Working effectively in Agile development environments. Specialized in building scalable microservices, and real time communication platforms with a focus on performance, reliability and distributed system with high availability.

EXPERTISE

- **Backend & Messaging:** Java Spring Boot, Erlang, Golang, Microservices, Ejabberd, Xmpp.
- **Frontend & Web:** Html, Javascript, React.js, Vite.
- **Messaging & Streaming:** Grpc Protobuf, RabbitMQ, Kafka.
- **Databases & Caching:** MySQL, SQLite, Redis, MongoDB, Cassandra.
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD, Jenkins, AWS, Oracle, Minio (on-premises), GCP.
- **Tools & LLM:** Jira, Grafana, Postman, Git, Bitbucket, Github, Psi, OpenCode, Claude.

EXPERIENCE

Contus Tech

Sep 2024 – June 2026

Designation: *Junior Software Development Engineer*

- Re-architected the authentication pipeline by replacing external Go/Python authentication services with **JWT-based authentication** backed by **Redis session validation**. Reduced authentication latency from **200ms–3s to 1–2ms**, enabled multi-session login through schema redesign, resolved database deadlocks, and introduced scheduled cleanup of expired JWT tokens.
- Eliminated the Java XMPP microservice by developing a native **RabbitMQ module in ejabberd** with asynchronous worker pools for publish/subscribe processing, reducing infrastructure overhead, removing inter-service network calls, and improving message throughput.
- Replaced the Cloud API Gateway proxy layer with highly available **Nginx** and **Apache** routing, removing single-service dependencies, minimizing internal REST communication, and improving request/response latency.
- Modernized the microservice architecture by migrating from **Spring MVC + Tomcat** to **Spring WebFlux + Netty** for non-blocking I/O and advocated adoption of **R2DBC** over traditional JDBC/JPA for reactive database access.
- Introduced **gRPC** with **Protocol Buffers** for internal service-to-service communication, significantly reducing serialization overhead and improving communication performance.
- Designed a horizontally scalable **Redis caching and persistence** architecture on Kubernetes to offload SQL traffic by caching user presence, blocking status, mute configurations, roster data, and group participants, substantially improving real-time messaging performance.
- Orchestrated the **message reactions** feature for one-to-one and group messaging using **ejabberd**. Implemented asynchronous database operations, offline synchronization using **Firestore Cloud Messaging (FCM)** and **Apple Push Notification service (APNs)**, and reliable persistence as a fallback mechanism.

- Rewrote the contact synchronization microservice from Java to **Go**, overcoming **JVM heap limitations** for large datasets. Reduced database load through Redis caching and introduced batched, parallel synchronization strategies to improve scalability.
- Optimized high-volume MySQL workloads through native query tuning, parameterized queries, composite indexing, and transactional consistency, significantly improving message-processing efficiency while ensuring reliable rollback behavior.
- Implemented end-to-end observability across the messaging platform by introducing tracing, structured logging, and performance metrics throughout ejabberd and backend microservices, enabling faster diagnosis of message delivery failures.
- Integrated the **Firestore Admin SDK** into the backend to validate **Firestore Authentication** ID tokens during user registration, ensuring secure user verification before account creation.
- Developed in-app purchase functionality for **Google Play** and the **Apple App Store**, supporting **recurring subscriptions** and **one-time purchases** to unlock anonymous communication features across mobile platforms.
- Worked extensively with real-time communication technologies including Ejabberd, RabbitMQ, Redis, Spring WebFlux, Go, gRPC, Protocol Buffers, Janus, SIP, LiveKit (WebRTC), and Centrifugo.

Innov Source — Tech Mahindra's client

Apr 2024 – Sep 2024

Designation: *FTTX Engineer*

- Developed a strong foundation in binary data representation and how digital information is transmitted across networks using **electromagnetic signals** and **fiber-optic communication** in cellular and Ethernet environments.
- Presented router fundamentals and **port forwarding** concepts to the team, improving understanding of network traffic routing and service accessibility.
- Gained hands-on understanding of how **public** and **private IP addresses**, NAT, and port mapping work together to route packets to specific devices and processes.

PRODUCT EXPERIENCE

Products I have dealt with:

- Core: **MirrorFly**, **Hike Messenger**.
- Forked: **Ejabberd XMPP**, **LiveKit WebRTC**, **Centrifugo TCP/WebSocket**.

CERTIFICATIONS

- **Spring Boot** — AmigosCode [view](#) Aug 2023 – Mar 2024
Hands-on course focused on building RESTful APIs in microservices.
- **Java: Spring** — QSpiders
Comprehensive training covering Core Java, OOP, collections, multithreading, exception handling.
- **MERN Stack** — EMC [view](#)
Focused on building modern web apps and integrating REST APIs using React, Node.js and mongoDB.

EDUCATION

Bachelor of Engineering — Computer Science and Engineering
Hindusthan Institute of Technology, CGPA: 7.8

Jun 2019 – May 2023